

First 90 Days Schedule

First 90 Days — Week-by-Week Starter Schedule (v10 on-ramp)

Covers: the ~16-week on-ramp from a standing start to your first portfolio capstone — **Phase 0 (fundamentals) → first job applications → Phase 1 (math) → Phase 2 + Capstone 1**. Dates assume a **mid-Aug 2026 start** at an aggressive-but-sustainable pace (~38-42 hrs/wk on working weeks, ~18-20 on consolidation weeks). The **self-tests are the true gates** — if one isn't passed, repeat the week; don't advance on shaky ground.

How to read this: each row is one week. The two consolidation weeks (6 and 12) are deliberate — **do not skip them**; they're what keep confusing terminology from compounding. Applications start in Week 3 and run as a light background track thereafter (your applied medical-imaging skill already makes you eligible — the learning improves your interview *performance*, not your eligibility).

The daily rhythm (once you're rolling)

- **Weekdays (~3-4 hrs/day):** 1 focused evening block. Once threads start (Week 10), the **first ~45 min is a thread** (rotate T → M → R), then phase work.
- **Weekend (~10-12 hrs/day):** 2-3 blocks with breaks. Heaviest lifting (new hard concepts) goes here, when you're fresh.
- **Every night (~15 min):** update the **living glossary** + clear your **Anki** queue (see the Confusion Buffer one-pager).
- **Confusion Buffer, always on:** intuition-first (visual → rigorous → derive), Feynman-gate each concept, and if stuck >30 min, switch teacher rather than grind.

Weeks 1-16

Wk	Dates (2026)	Focus & key resources	Hrs	Milestones / flags
1	Aug 18-23	Phase 0.1 Statistics — StatQuest stats playlist. Set up your glossary doc + Anki (day 1).	~32	Ramp-up week. Start the glossary habit immediately
2	Aug 24-30	Phase 0.2 Classic ML (bias-variance, ROC/AUC, regularization, trees/boosting, SVM/kernels, PCA) — StatQuest	~40	Anki: add ML-term cards nightly
3	Aug 31-Sep 6	Phase 0.2 finish + 0.3 DL fundamentals (StatQuest NN + 3B1B NN) + 0.4 articulation cheat-sheet. Phase 0 self-test (Feynman gate).	~40	► Phase 0 done → START APPLYING (Tier 0). Refresh CV/LinkedIn; first applications to Qure.ai, SigTuple, GE HealthCare, NVIDIA-India
4	Sep 7-13	Phase 1.1 Linear algebra — 3B1B LA (all, intuition first) → MIT 18.06 targeted lectures	~40	Triangulate SVD/eigen if it's slippery
5	Sep 14-20	Phase 1.1 continue — SVD & eigendecomposition deep; 18.065 SVD/PCA/low-rank sections; derive SVD from eigendecomposition on paper	~40	Whiteboard: $A^T A$ is PSD; rank-r approximation
6	Sep 21-27	CONSOLIDATION — no new material. Re-derive LA, glossary review, Anki catch-up. Keep applications ticking.	~18	⚙ Do not skip. Re-derive, don't re-watch
7	Sep 28-Oct 4	Phase 1.1 finish + 1.2 Calculus + matrix calculus — 3B1B Calc; Parr & Howard matrix-calculus	~40	Derive backprop through a linear layer w/ matrix calc
8	Oct 5-11	Phase 1.3 Probability & statistics — Harvard Stat 110 (targeted); Blitzstein & Hwang book	~40	Whiteboard: MLE for a Gaussian; conditional expectation as projection
9	Oct 12-18	Phase 1.3 finish + 1.4 Karpathy Zero-to-Hero start (micrograd)	~40	Re-implement micrograd from a blank file
10	Oct 19-25	Phase 1.4 bulk — makemore + attention build; re-implement each without looking at his code	~42	► Phase 1 self-test (backprop, micrograd, attention + $\sqrt{d_k}$) → THREADS T, M, R BEGIN
11	Oct 26-Nov 1	Phase 1.4 finish (GPT build) + Phase 2.1 imaging physics (Elster MRI Q&A; fluorescence primer). Threads now running (~1.5 hrs/day).	~42	Threads absorb ~10 hrs/wk from here on
12	Nov 2-8	CONSOLIDATION — re-derive Phase-1 math + attention; Anki; re-implement backprop + attention from blank file. Threads continue light.	~20	⚙ Do not skip
13	Nov 9-15	Phase 2.2 CNNs + conv math + U-Net family + losses — CS231n; conv-arithmetic; U-Net/nnU-Net papers. Implement	~4	Whiteboard U-Net with tensor shapes

		Dice/Focal/Tversky from scratch (Thread M).	2	
14	Nov 16–22	Phase 2.3 nnU-Net/MONAI packaging + 2.4 transformer segmentation (SAM/MedSAM/MedNeXt). Set up Capstone 1 (pick dataset, repo, plan).	~42	Choose Capstone 1 dataset (BraTS / Decathlon / LIVECell)
15	Nov 23–29	Phase 2.5 (3D, uncertainty, evaluation) + Capstone 1 build — preprocessing + first architecture + first ablation	~42	Metrics Reloaded for honest evaluation
16	Nov 30–Dec 6	Capstone 1 finish — architecture + loss ablations, uncertainty + failure-mode analysis, 3-page write-up, clean public repo	~42	► Capstone 1 done → Tier 0/1 interview-ready. Ramp applications + ask for referrals; Thread R: turn the write-up toward a MIDL/MICCAI-workshop short paper

Totals: ~610 focused hours over 16 weeks (~38 hrs/wk average, including two consolidation weeks). This puts you Tier-0/1 interview-ready by early December, with applications already in flight since Week 3 — exactly the incremental on-ramp the plan is built around.

The applications sub-track (Weeks 3 → ongoing, ~2 hrs/wk)

- Week 3:** refresh CV + LinkedIn (lead with medical-imaging segmentation + production ML; frame CV-beyond-medical honestly as growing). Draft a 1-paragraph "why me for digital biology" narrative.
- Weeks 3–8:** apply to **Tier 0** (Qure.ai, SigTuple, GE HealthCare Bengaluru, NVIDIA-India, MSR India). ~3–5 quality applications/week > mass spraying. Track in a simple sheet.
- Weeks 9–16:** as Capstone 1 takes shape, add it to your CV/portfolio, request warm referrals, and start Tier-0 interview loops. Treat early loops as **paid calibration** — every rejection is interview-format data.
- From Week 16:** Capstone 1 live → strengthen every application; begin stretching toward Google-India and (once Capstone 2 lands ~Jan) Tier 1.

Reminder (COI): public data and generic methods only; personal hardware/accounts; clear anything public through Lilly's review. Nothing from Lilly or the QpiAI engagement.

What "done" looks like at the two gates in this window

- Phase 0 self-test (Week 3):** explain ~15 fundamentals out loud, plain language, no notes — p-value, CI, CLT, MLE, bias-variance, ROC/AUC vs PR, why L1→sparsity, backprop, BatchNorm. *This is your interview-articulation warm-up — pass it before applying.*
- Phase 1 self-test (Week 10):** hand-derive backprop for a 2-layer MLP; implement micrograd from memory; explain why BatchNorm helps and what breaks without it; whiteboard scaled-dot-product attention and derive the $\sqrt{d_k}$ scaling. *Threads only start once you pass — they reinforce a foundation, not a gap.*

If either gate isn't clean, add a repeat/consolidation week. Stretching the timeline a week beats carrying a hole into Phase 2.